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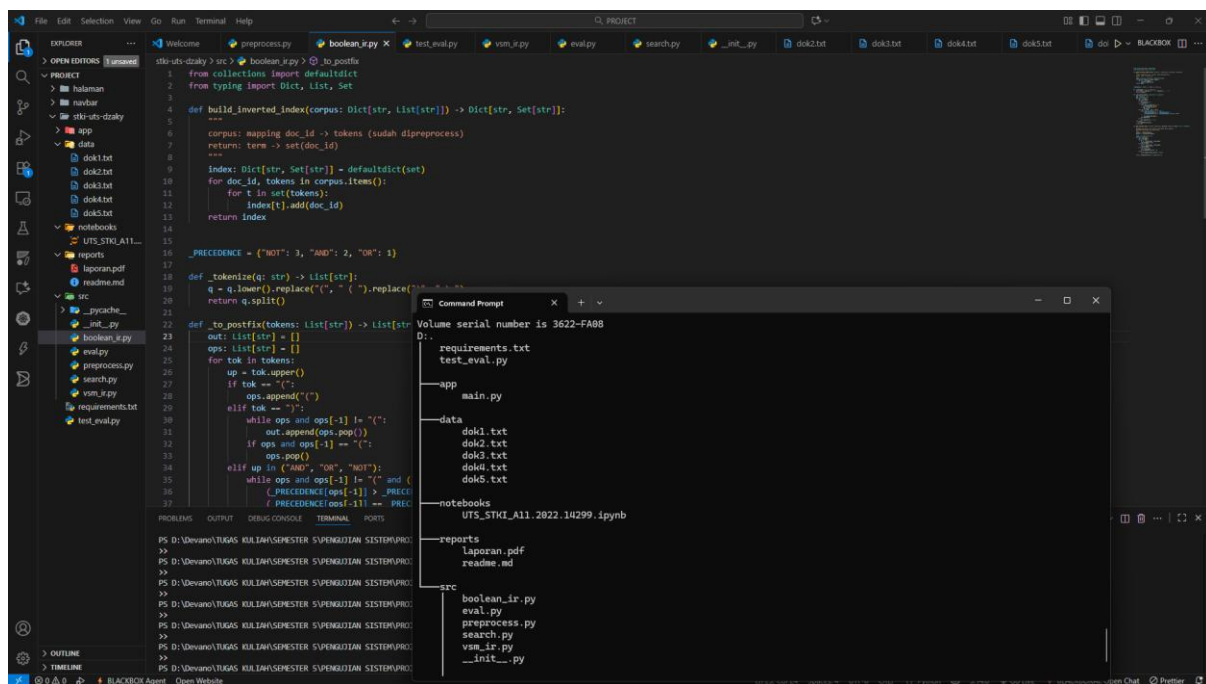
Kelompok : A11.4703

UJIAN TENGAH SEMESTER

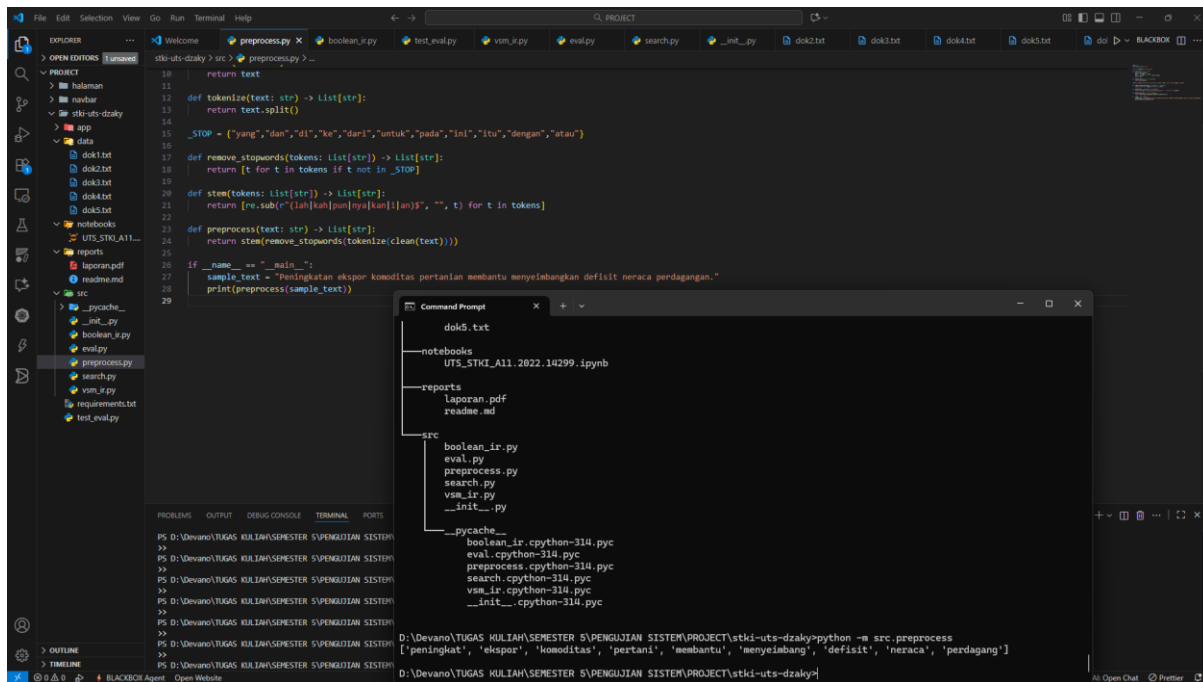
Sistem Temu Kembali Informasi

1. “Proyek ini bertujuan membangun sistem temu kembali informasi sederhana dengan korpus 10 dokumen teks. Sistem mengimplementasikan model Boolean dan Vector Space Model (VSM) untuk mendukung pencarian berbasis kata kunci. Proyek ini merealisasikan Sub-CPMK 10.1.1–10.1.4 melalui konsep teori, preprocessing, perancangan retrieval model, dan evaluasi hasil pencarian.”

STRUKTUR SISTEM



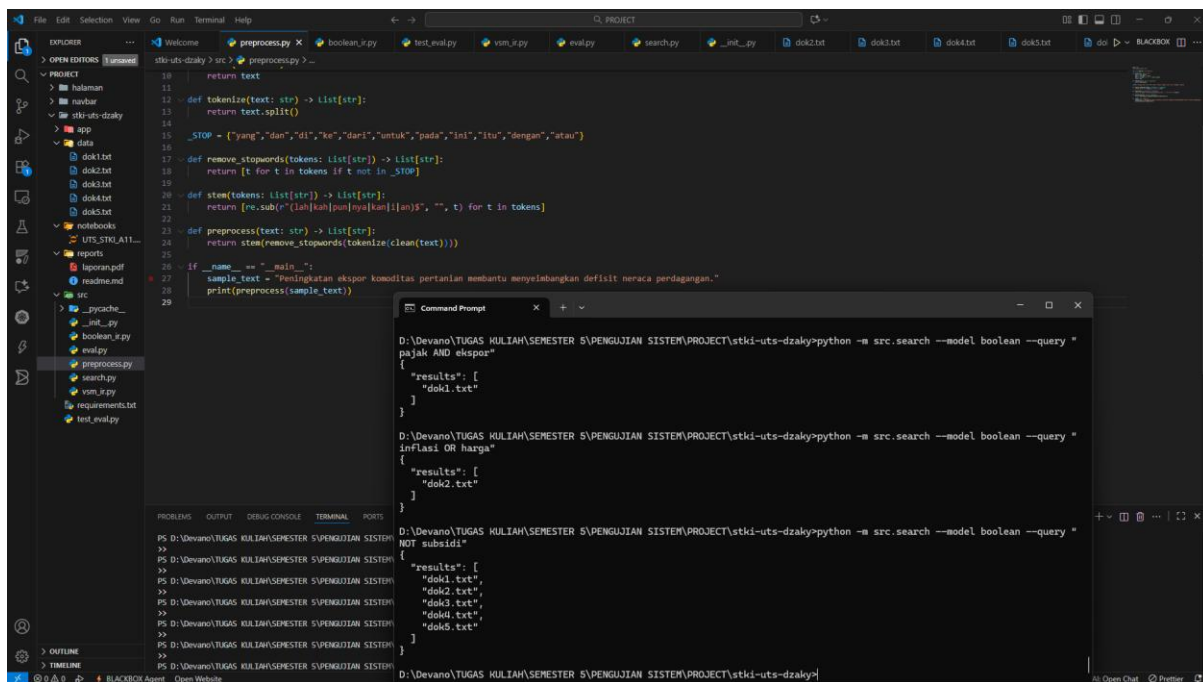
2. PERPROCESSING



```
10: return text
11:
12: def tokenize(text: str) -> List[str]:
13:     return text.split()
14:
15: _STOP = ("yang", "dan", "di", "ke", "dari", "untuk", "pada", "ini", "itu", "dengan", "atau")
16:
17: def remove_stopwords(tokens: List[str]) -> List[str]:
18:     return [t for t in tokens if t not in _STOP]
19:
20: def stem(tokens: List[str]) -> List[str]:
21:     return [re.sub(r'([a-z][a-z][a-z][a-z])$', '', t) for t in tokens]
22:
23: def preprocess(text: str) -> List[str]:
24:     return stem(remove_stopwords(tokenize(clean(text))))
25:
26: if __name__ == "__main__":
27:     sample_text = "Peningkatan ekspor komoditas pertanian membantu menyeimbangkan defisit neraca perdagangan."
28:     print(preprocess(sample_text))
29:
```

```
PS D:\Devano\TUGAS KULIAH\SEMESTER 5\PENGUJIAN SISTEM> python -m src.preprocessing
['peningkat', 'ekspor', 'komoditas', 'pertani', 'membantu', 'menyeimbang', 'defisit', 'neraca', 'perdagangan']
```

3. BOOLEAN RETRIEVAL MODEL



```
10: return text
11:
12: def tokenize(text: str) -> List[str]:
13:     return text.split()
14:
15: _STOP = ("yang", "dan", "di", "ke", "dari", "untuk", "pada", "ini", "itu", "dengan", "atau")
16:
17: def remove_stopwords(tokens: List[str]) -> List[str]:
18:     return [t for t in tokens if t not in _STOP]
19:
20: def stem(tokens: List[str]) -> List[str]:
21:     return [re.sub(r'([a-z][a-z][a-z][a-z])$', '', t) for t in tokens]
22:
23: def preprocess(text: str) -> List[str]:
24:     return stem(remove_stopwords(tokenize(clean(text))))
25:
26: if __name__ == "__main__":
27:     sample_text = "Peningkatan ekspor komoditas pertanian membantu menyeimbangkan defisit neraca perdagangan."
28:     print(preprocess(sample_text))
29:
```

```
D:\Devano\TUGAS KULIAH\SEMESTER 5\PENGUJIAN SISTEM\PROJECT\stki-uts-dzaky>python -m src.search --model boolean --query "
pajak AND ekspor"
{
  "results": [
    "dok1.txt"
  ]
}
D:\Devano\TUGAS KULIAH\SEMESTER 5\PENGUJIAN SISTEM\PROJECT\stki-uts-dzaky>python -m src.search --model boolean --query "
inflasi OR harga"
{
  "results": [
    "dok2.txt"
  ]
}
D:\Devano\TUGAS KULIAH\SEMESTER 5\PENGUJIAN SISTEM\PROJECT\stki-uts-dzaky>python -m src.search --model boolean --query "
NOT subsidi"
{
  "results": [
    "dok1.txt",
    "dok2.txt",
    "dok3.txt",
    "dok4.txt",
    "dok5.txt"
  ]
}
```

4. VECTOR SPACE MODEL & RANKING

```
1 Nilai tukar rupiah menguat karena peningkatan ekspor dan penurunan impor.  
2 Stabilitas ekonomi nasional sangat bergantung pada kebijakan fiskal dan moneter.
```

```
"results": [  
    "dok2.txt"  
]  
}  
  
D:\Devano\TUGAS KULIAH\SEMESTER 5\PENGUJIAN SISTEM\PROJECT\stki-uts-dzaky>python -m src.search --model boolean --query "NOT subsidi"  
{  
    "results": [  
        "dok1.txt",  
        "dok2.txt",  
        "dok3.txt",  
        "dok4.txt",  
        "dok5.txt"  
    ]  
}  
  
D:\Devano\TUGAS KULIAH\SEMESTER 5\PENGUJIAN SISTEM\PROJECT\stki-uts-dzaky>python -m src.search --model vsm --query "pajak ekspor harga"  
{  
    "results": [  
        "dok1.txt",  
        "dok3.txt",  
        "dok2.txt",  
        "dok4.txt",  
        "dok5.txt"  
    ]  
}
```

5. TERM WEIGHTING, SEARCH ENGINE, DAN EVALUASI

```
1 from collections import defaultdict  
2 from typing import Dict, List, Set  
3  
4 def build_inverted_index(corpus: Dict[str, List[str]]) -> Dict[str, Set[str]]:  
5     """  
6     corpus: mapping doc_id -> tokens (sudah dipreprocess)  
7     return: term -> set(doc_id)  
8     """  
9     index: Dict[str, Set[str]] = defaultdict(set)  
10    for doc_id, tokens in corpus.items():  
11        for t in set(tokens):  
12            index[t].add(doc_id)  
13    return index  
14  
15 _PRECEDENCE = ("NOT": 3, "AND": 2, "OR": 1)  
16  
17 def tokenize(s: str) -> List[str]:  
18    q = s.lower().replace("(", " ( ").replace(")", " ) ")  
19    return q.split()  
20  
21 def to_postfix(tokens: List[str]) -> List[str]:  
22    out: List[str] = []  
23    ops: List[str] = []  
24    for tok in tokens:  
25        up = tok.upper()  
26        if tok == "(":  
27            ops.append("(")  
28        elif tok == ")":  
29            while ops and ops[-1] != "(":  
30                out.append(ops.pop())  
31            ops.pop()  
32        elif up in ("AND", "OR", "NOT"):  
33            ops.append(up)  
34        elif up in ("AND", "OR", "NOT"):  
35            while ops and ops[-1] != "(" and  
36                _PRECEDENCE[ops[-1]] > _PRECEDENCE[up]:  
37                out.append(ops.pop())  
38    out.append(ops.pop())  
39    return out
```

```
"dok5.txt"  
}  
  
D:\Devano\TUGAS KULIAH\SEMESTER 5\PENGUJIAN SISTEM\PROJECT\stki-uts-dzaky>python test_eval.py  
Precision: 0.5  
Recall: 0.5  
F1-score: 0.5  
  
D:\Devano\TUGAS KULIAH\SEMESTER 5\PENGUJIAN SISTEM\PROJECT\stki-uts-dzaky>python -m src.search --model boolean --query "pajak AND ekspor"  
{  
    "results": [  
        "dok1.txt"  
    ]  
}  
  
D:\Devano\TUGAS KULIAH\SEMESTER 5\PENGUJIAN SISTEM\PROJECT\stki-uts-dzaky>python -m src.search --model vsm --query "pajak ekspor"  
{  
    "results": [  
        "dok1.txt",  
        "dok3.txt",  
        "dok2.txt",  
        "dok4.txt",  
        "dok5.txt",  
        "dok2.txt"  
    ]  
}
```

DATA

```
File Edit Selection View Go Run Terminal Help
PROJECT
> OPEN EDITORS 2 unsaved
stku-uts-dzaky > data > dok1.txt
1 Pemerintah meningkatkan pajak ekspor untuk menstabilkan nilai tukar rupiah.
2 Kebijakan ini bertujuan mengurangi impor barang mewah dan memperkuat cadangan devisa.
```

```
File Edit Selection View Go Run Terminal Help
PROJECT
> OPEN EDITORS 2 unsaved
stku-uts-dzaky > data > dok2.txt
1 Inflasi meningkat akibat kenaikan harga bahan bakar dan bahan pangan.
2 Bank Indonesia menyesuaikan suku bunga untuk menekan laju inflasi.
```

```
File Edit Selection View Go Run Terminal Help
PROJECT
> OPEN EDITORS 2 unsaved
stku-uts-dzaky > data > dok3.txt
1 Peningkatan ekspor komoditas pertanian membantu menyeimbangkan defisit neraca perdagangan.
2 Dukungan pemerintah pada UMKM berkontribusi terhadap pertumbuhan ekspor nonmigas.
```

```
File Edit Selection View Go Run Terminal Help
PROJECT
> OPEN EDITORS 2 unsaved
stku-uts-dzaky > data > dok4.txt
1 Pajak penghasilan dinaikkan untuk meningkatkan pendapatan negara.
2 Reformasi perpajakan dilakukan agar sistem lebih adil dan efisien.
```

```
File Edit Selection View Go Run Terminal Help
PROJECT
> OPEN EDITORS 2 unsaved
stku-uts-dzaky > data > dok5.txt
1 Nilai tukar rupiah menguat karena peningkatan ekspor dan penurunan impor.
2 Stabilitas ekonomi nasional sangat bergantung pada kebijakan fiskal dan moneter.
```

NOTEBOOKS

